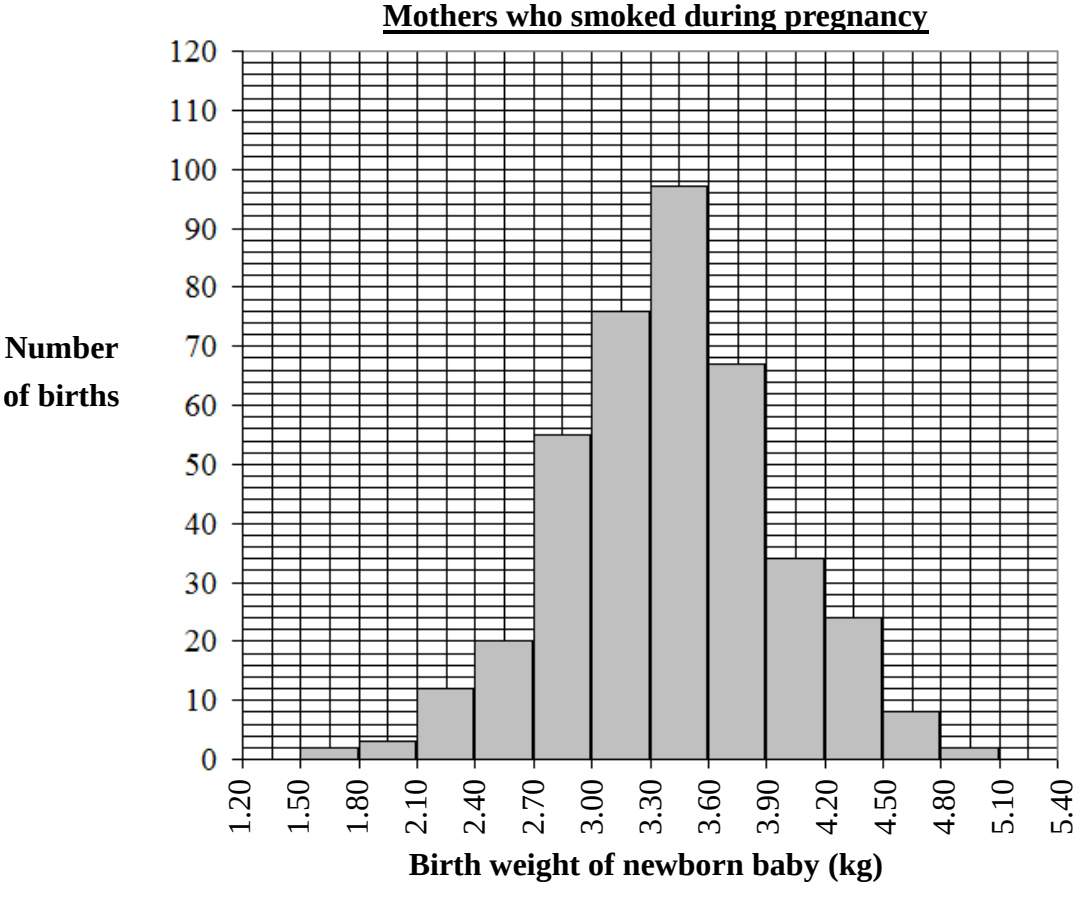
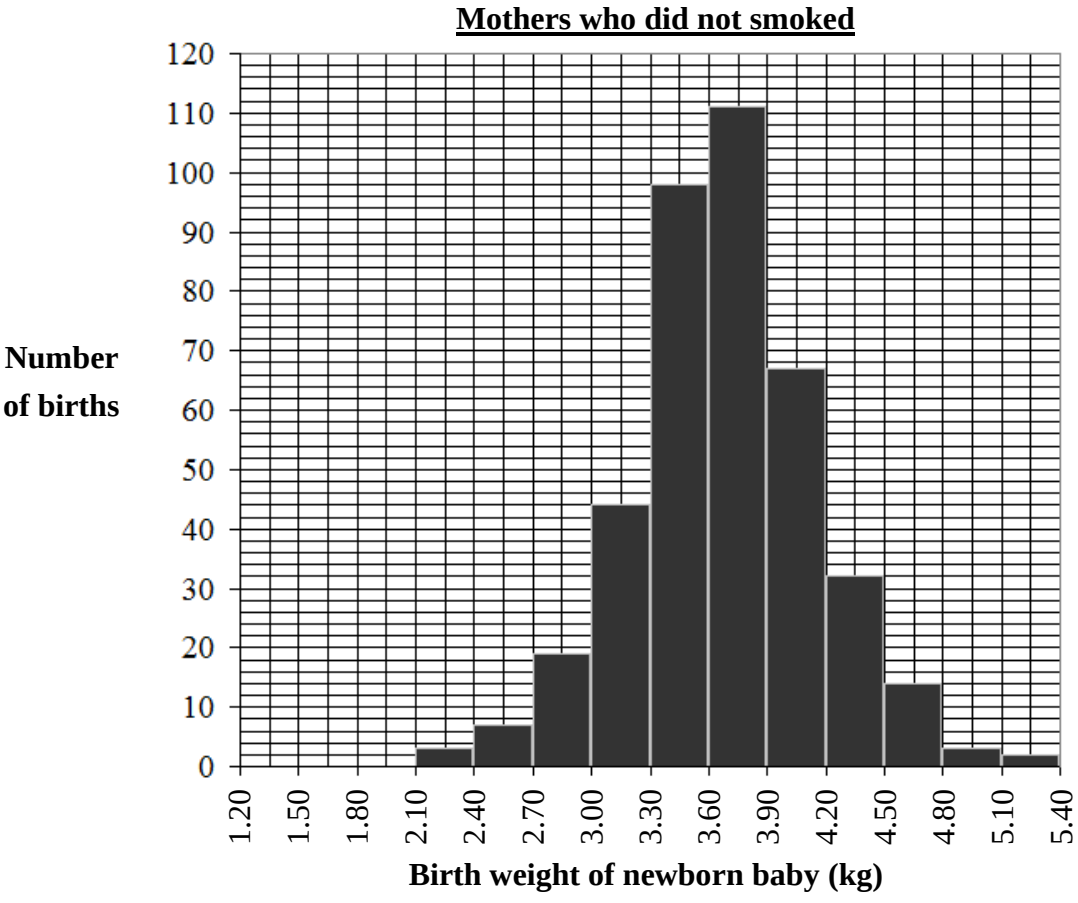
Structured questions

|!|BCELB0140000001|!| ★ **SI** **STSE**

Smoking during pregnancy is hazardous to the foetus. Researchers collected data about the birth weights of newborn babies from mothers who smoked during pregnancy and mothers who did not smoke. The charts below show the results.





(a) Cigarette smoke contains a lot of harmful substances, such as nicotine. Construct a flowchart to show the route by which nicotine in cigarette smoke travels from the mother's lungs to the foetus. (Include only the major organs and blood vessels involved) (2 marks)

(b) Doctors consider a baby to be underweight if the baby weighs less than 2.40 kg. Based on the charts given, find out how many babies from the mothers who smoked and who did not smoke were underweight. (2 marks)

(c) Write ***one*** conclusion that can be drawn based on the charts. (1 mark)

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(d) Describe how carbon monoxide in cigarette smoke may cause harm to a foetus.
(3 marks)

##

(a) (Mother's lungs →) pulmonary vein → heart → aorta → artery to the uterus → placenta → umbilical vein (→ foetus)
Correct sequence (1)
Correct format (1)

(b) From the mothers who smoked: 17 babies (1)
From the mothers who did not smoke: 3 babies (1)

(c) The most frequent birth weight of babies from the mothers who did not smoke is higher than that from the mothers who smoked. /
There are more underweight babies from the mothers who smoked compared with the mothers who did not smoke. /
The babies born from the mothers who smoked generally have a lower birth weight compared with the babies from the mothers who did not smoke.
(Or other reasonable answers) (1)

(d) Carbon monoxide binds irreversibly with haemoglobin in red blood cells. (1)
This reduces the oxygen-carrying capacity of red blood cells. (1)
Oxygen supply to the foetus decreases. (1)

(8 marks)

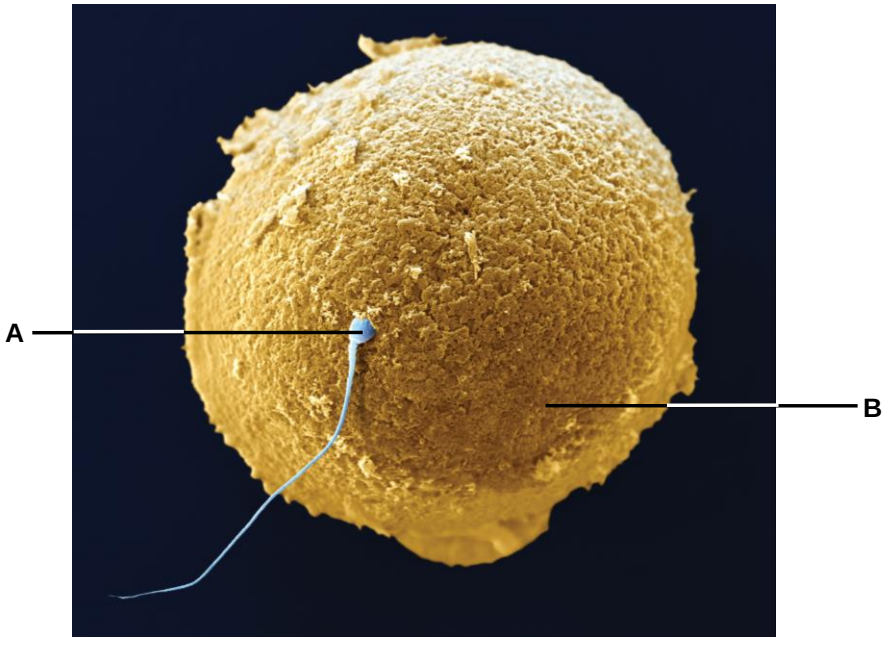
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
	Presenting information	1
(b)	Analysing data	2
(c)	Analysing data	1
(d)	Understanding basic concepts	3

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|!|BCELB0140000002|!| ★

The photomicrograph below shows two human gametes.



Human gametes (×700)

(a) With reference to the photomicrograph, describe ***two*** differences between gamete A and gamete B. For each difference, explain how it helps in the process of reproduction. (4 marks)

(b) (i) Name the process shown in the photomicrograph. (1 mark)

(ii) Describe the change in gamete B at the moment when it makes contact with gamete A. State the significance of this change. (2 marks)

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(iii) A woman would like to carry out tubal ligation to prevent the process shown in the photomicrograph. She wonders whether the operation will lead to the stop of menstruation. If you were her doctor, explain whether tubal ligation would affect menstruation. (4 marks)

##

(a) Gamete A has tail while gamete B does not. (1)
The tail helps gamete A to swim towards gamete B for fertilization to occur. (1)
The size of gamete B is larger than gamete A. (1)
This allows gamete B to store more food reserve to nourish the embryo. (1)

(b) (i) Fertilization (1)

(ii) The jelly coat of gamete B thickens and hardens. (1)
This ensures gamete B is only fertilized by one gamete A, ensuring the zygote formed is diploid. (1)

(iii) Menstruation would not be affected. (1)
Menstruation is controlled by female sex hormones secreted by the ovaries. (1)
After tubal ligation, female sex hormones can still be secreted by the ovaries and transported to the target organs, such as uterine lining, by the blood. (1)
Under the actions of the female sex hormones, the uterine lining will get thicker and break down later. (1)

(11 marks)

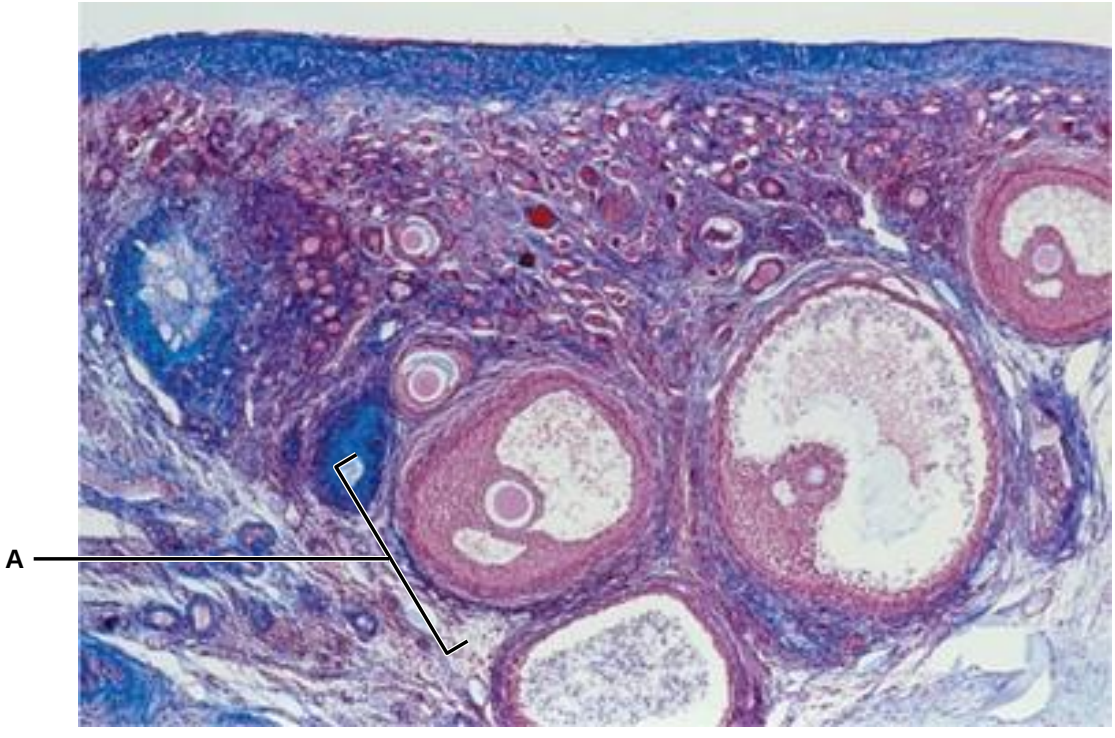
##

	Skill assessed	Marks
(a)	Analysing data	4
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2
(b)(iii)	Applying concepts	4

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|!|BCELB0140000003|!| ★

The photomicrograph below shows a section of human ovary.



A section of human ovary (×200)

(a) State and describe the process which takes place when structure A becomes mature. (3 marks)

(b) (i) Name the structure in which structure A will develop into after the process mentioned in (a). (1 mark)

(ii) State ***one*** hormone secreted by this structure and its function. (2 marks)

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(c) Explain whether the removal ovaries from a sexually immature girl will affect her development of secondary sexual characteristics. (3 marks)

##

(a) Ovulation (1)
Structure A bursts open, (1)
releasing ovum to the opening of the oviduct. (1)

(b) (i) Structure A will develop into yellow body. (1)
(ii) Oestrogen / progestogen (1)
It simulates the thickening of the uterine lining to prepare for implantation. (1)

(c) After the removal of ovaries, the girl can no longer produce female sex hormones. (1)
Female sex hormones are essential for the development of secondary sexual characteristics. (1)
Therefore, the development of secondary sex characteristics of the girl is prevented. (1)

(9 marks)

##

	Skill assessed	Marks
(a)	Analysing data	3
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	2
(c)	Applying concepts	3

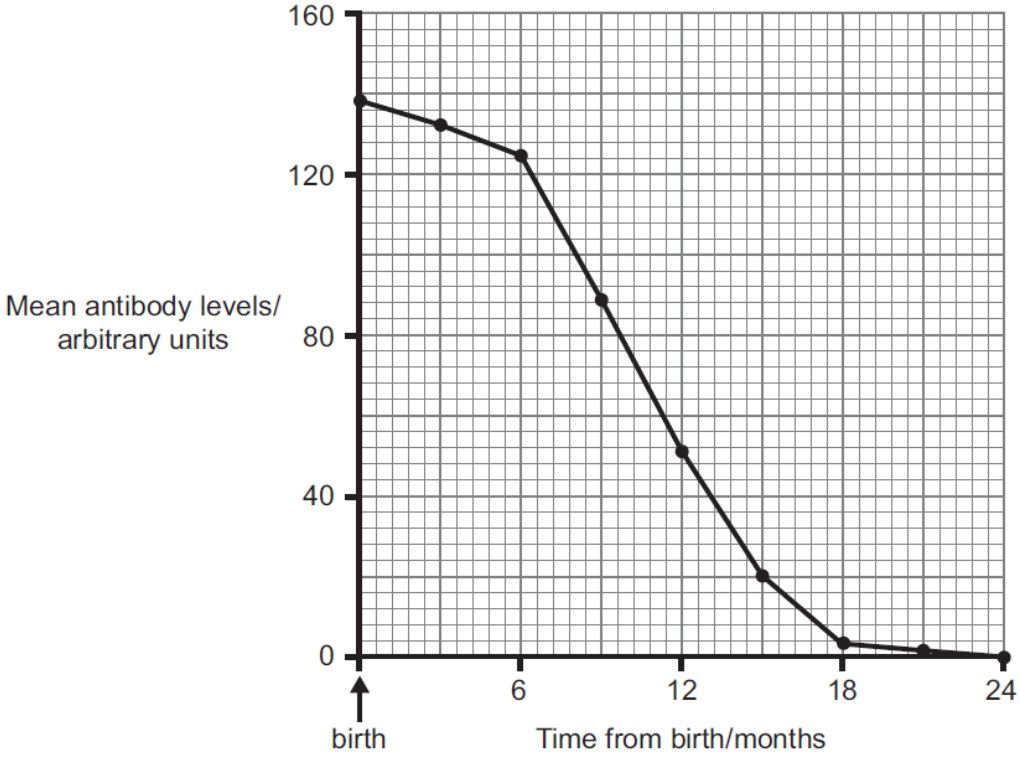
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|!|BCELB0140000004|!| ★ **Cross topic**

CCEA GCE Biology Assessment Unit A2 1 25 Jan 2010 Q2

Many people have an immunity to measles as a result of either infection or vaccination. Measles vaccination starts with infants at 15 months.

The measles antibody levels were measured in a group of babies during the first 24 months of life. The results for mean antibody levels are shown in the graph below.



(a) Explain the high levels of the measles antibodies in babies at birth and for the first six months. (2 marks)

(b) Explain the low levels of the antibodies in babies after 12 months. (2 marks)

(c) Explain why a measles vaccine would not be successful if given within the first six months. (1 mark)

##

(a) Any ***two*** of the following : (2)
- Foetuses receive the antibodies passively from the mother (1)
- via the placenta (1)
- Babies receive the antibodies in breast milk (1)

(b) Many babies are no longer receiving the antibodies in the milk / antibodies are not replaced (1)
Degradation of the antibodies / antibodies used up (1)

(c) The antibodies presence in some babies would destroy the vaccine / high levels of the antibodies inhibit the production of plasma and memory cells (1)

(5 marks)

##

	Skill assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	2
(c)	Applying concepts	1

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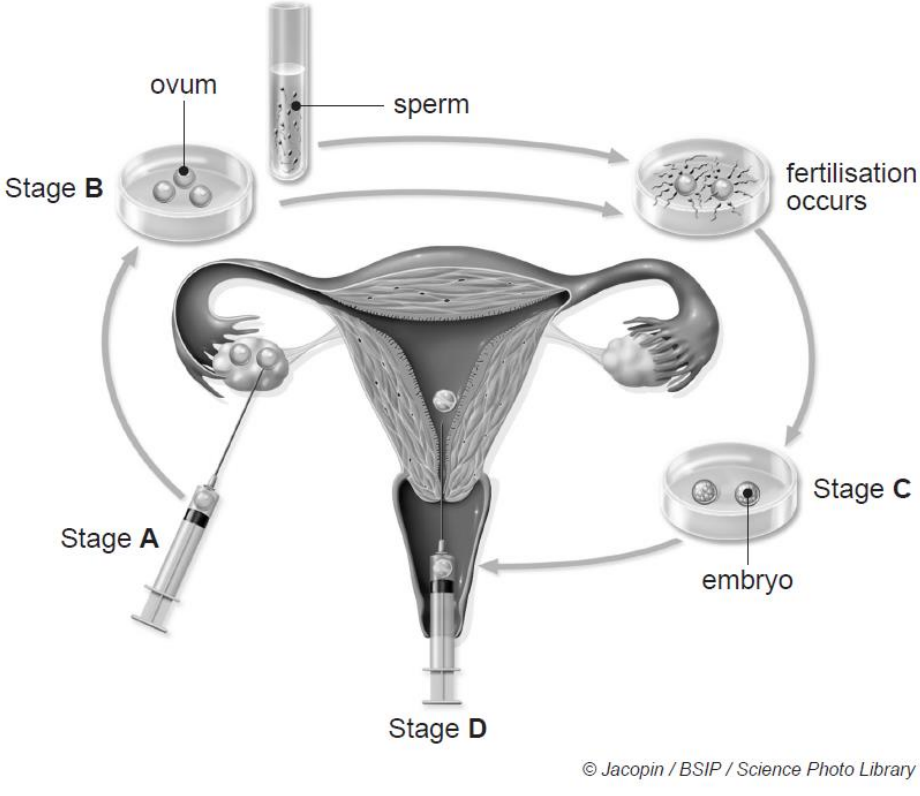
|!|BCELC0140000005|!| **SI** **STSE**

CCEA GCSE Biology Unit 2 Higher Tier 7 Jun 2019 Q6

Some couples experience infertility problems when trying to conceive a child.

(a) Give ***three*** causes of infertility. (3 marks)

The diagram shows the stages of *in vitro* fertilization (IVF) which can be used to help couples to conceive a child.



© Jacopin / BSIP / Science Photo Library

(b) During stage A, several ova are removed from the woman's ovary.

(i) Describe how the woman's ovaries are stimulated to produce several ova at the same time. (1 mark)

(ii) Explain why several ova are removed from the woman's ovary. (1 mark)

(c) Following stage D, implantation occurs.

(i) What is implantation? (2 marks)

(ii) Name the process which takes place in the embryo, following implantation, to produce a variety of tissues and organs. (1 mark)

##

(a) Any ***three*** of the following: (3)
Ovaries may not release / produce an ovum (1)
Oviducts blocked (1)
Uterine / vagina environment hostile (1)
Man has a low sperm count (1)
Sperm may have low motility / unhealthy (1)
(Or other reasonable answers)

(b) (i) Use fertility drugs / hormones (1)

(ii) Increases the chances that ova may be fertilized / embryo is formed (1)

(c) (i) The embryo embeds itself (1)
in the uterine lining (1)

(ii) Differentiation (1)

(8 marks)

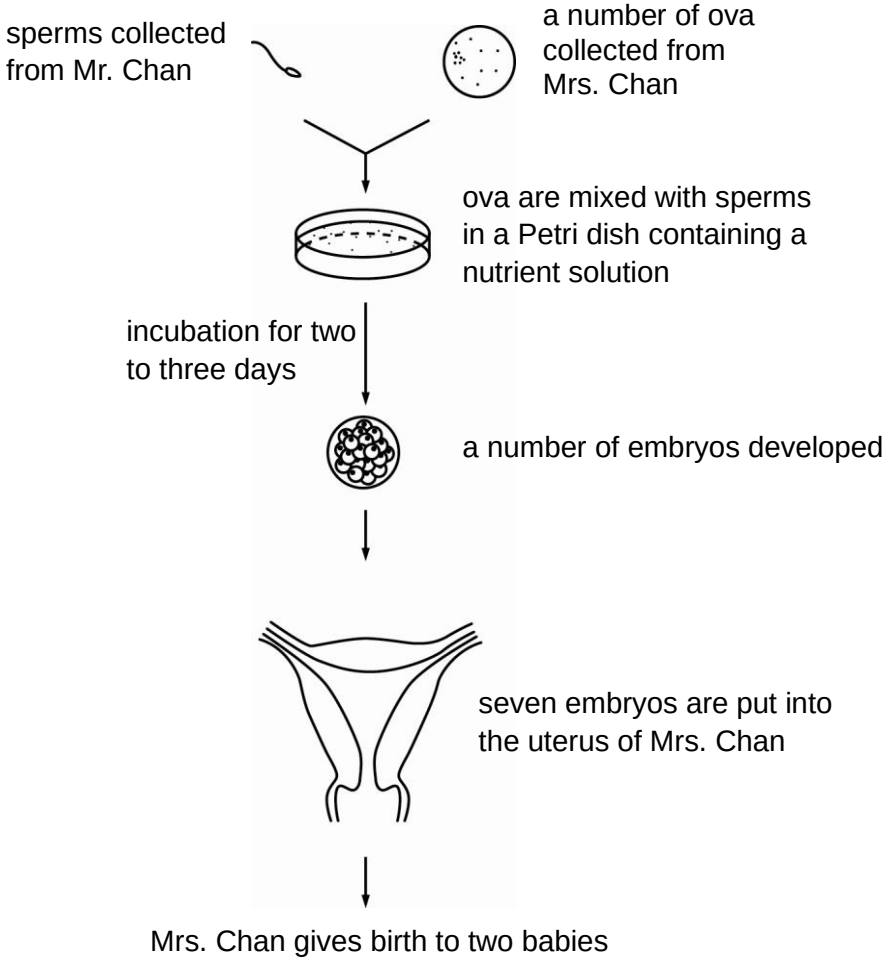
##

	Skill assessed	Marks
(a)	Understanding basic concepts	3
(b)(i)	Understanding basic concepts	1
(b)(ii)	Applying concepts	1
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	1

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|!|BCELB0140000006|!| ★ **SI** **STSE**

Mr. and Mrs. Chan can produce gametes but they are still unable to give birth to children. A doctor advised them to have an in vitro fertilization (IVF) treatment. The following diagram shows the procedures of IVF.



(a) Suggest ***one*** possible cause of infertility for Mr. and Mrs. Chan respectively. (2 marks)

_____

_____

(b) Describe how an embryo is formed from an ovum in the Petri dish. (2 marks)

_____

_____

(c) Of the seven embryos put into the uterus, only two embryos have successfully developed.

(i) Explain why some embryos cannot develop in the uterus? (1 mark)

(ii) Is it possible that the two embryos have different blood groups? Explain your answer. (3 marks)

(d) Suggest ***one*** social / ethical issue that may arise from the application of IVF in giving birth to children. (1 mark)

##

(a) For Mr. Chan:
low sperm count / poor sperm mobility / sperms with structural defects (1)
For Mrs. Chan:
blocked oviduct / unsuccessful implantation / no ovulation (1)

(b) An ovum is fertilized by a sperm to form a zygote. (1)
The zygote divides by mitotic cell division to form an embryo. (1)

(c) (i) Some embryos fail to implant in the uterus. (1)

(ii) As the embryos may develop from the fertilization of two ova with two different sperms. (1)
The gametes may carry different alleles for the blood group. (1)
Therefore, the embryos may have different blood groups. (1)

(d) Any ***one*** of the following: (1)
- Many embryos are sacrificed in this process. (1)
- It may lead to the selection of sex of baby and affect the gender ratio in the population. (1)

(Or other reasonable answers)

(9 marks)

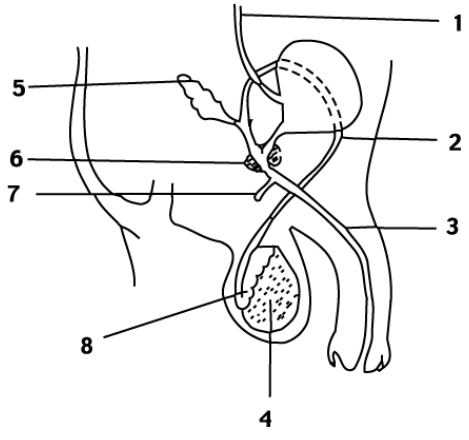
##

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	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Applying concepts	3
(d)	Critical thinking	1

|!|BCELC0140000007|!|

The diagram below shows the male reproductive system.



(a) (i) Which structures are involved in the production of seminal fluid? (2 marks)

(ii) Give ***two*** functions of the seminal fluid to sperms. (2 marks)

(b) Identify structure 8 and state its function. (2 marks)

(c) (i) Name ***one*** contraceptive method that involves structure 2. (1 mark)

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(ii) Explain how this contraceptive method can achieve contraception. (1 mark)

_____

_____

##

(a) (i) 4, 5, 6, 7 (deduct 1 mark for 1 wrong answer) (2)

(ii) Any ***two*** of the following: (2)

- It contains nutrients which nourish the sperms. (1)
- It neutralizes the acidic secretions of the female's vagina. (1)
- It provides a liquid medium for sperms to swim in. (1)

(b) Epididymis (1)

It stores the sperms temporarily. (1)

(c) (i) Vasectomy (1)

(ii) It prevents sperms from passing through the sperm ducts and so the semen does not contain sperms. (1)

(8 marks)

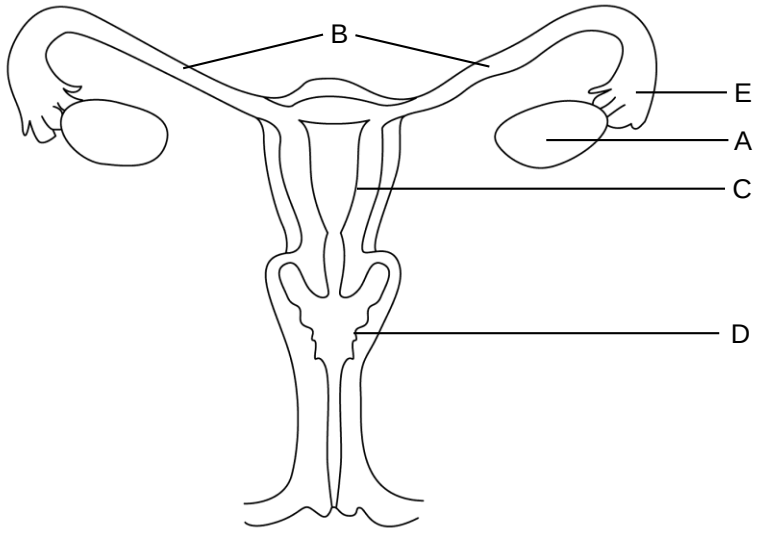
##

	Skill assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Understanding basic concepts	2
(b)	Analysing data	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	1

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|!|BCELC0140000008|!|

The diagram below shows the human female reproductive system.



(a) Name structure A and state ***two*** of its functions. (3 marks)

(b) Describe how structures B and C are adapted to their functions. (4 marks)

(c) Describe ***two*** contraceptive methods that involve the structures above. (4 marks)

##

(a) Ovary (1)

Any ***two*** of the following: (2)

- produce ova (1)
- produce female sex hormones (1)

(Or other reasonable answers)

(b) Structure B (oviduct) is a muscular tube lined with ciliated epithelium. (1)

The beating actions of cilia and muscular contractions of the oviduct wall help move an ovum along the oviduct. (1)

Structure C (uterus) is a thick-walled muscular organ. (1)

It provides protection and a stable environment for the development of the embryo. (1)

(c) Any ***two*** sets of the following: (4)

1. tubal ligation: structures B are tied and cut (1)
   It prevents sperms from reaching the ovum. (1)
2. Intrauterine device placed in structure C (1)
   It prevents the implantation of an embryo. (1)
3. Diaphragm placed at the top of structure D (1)
   It prevents sperms from reaching the ovum. (1)

(11 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	4
(c)	Understanding basic concepts	4

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|!|BCELB0140000009|!| ★ **Cross topic**

The following two diagrams show a flower (***Diagram A***) and the human female reproductive system (***Diagram B***).

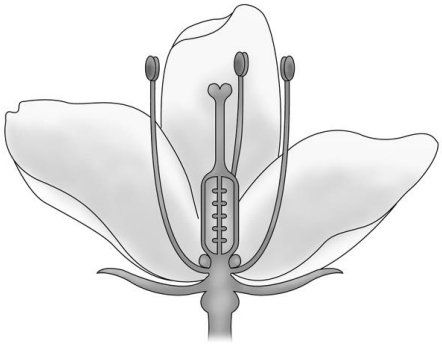


Diagram A

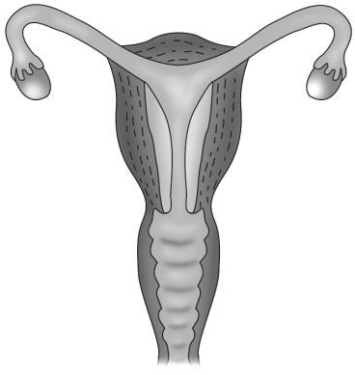


Diagram B

(a) Describe how the male gametes meet and fertilize the female gametes in

(i) flowering plants; (3 marks)

(ii) humans. (3 marks)

(b) State ***two*** differences between the reproduction of flowering plants and humans. (2 marks)

##

(a) (i) In flowering plants, the pollen grain develops a pollen tube which grows down the stigma. (1)

The pollen tube then enters the micropyle and releases the male gamete. (1)

The female gamete (ovum) inside the ovule is then fertilized by the male gamete. (1)

(ii) In humans, copulation takes place / male's penis is inserted into the vagina of the female. (1)

Sperms are discharged into the vagina after ejaculation. (1)

The sperms swim up the vagina and fertilize the ovum in the oviduct. (1)

(b) Any ***two*** of the following comparisons: (1 mark each) (2)

Reproduction in flowering plant	**Reproduction in human**
- Male gametes are transferred with the help of external agents.	- Male gametes are transferred directly to female in semen.
- Female gametes remain in the ovary.	- Female gametes are released from the ovary.
- Most of the embryo develops outside the parents.	- Embryo develops inside the parents.

(Or other reasonable answers)

(8 marks)

##

	Skill assessed	**Marks**
(a)(i)	Understanding basic concepts	3
(a)(ii)	Understanding basic concepts	3
(b)	Applying concepts	2

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|!|BCELB0140000010|!| ★ **STSE**

Read the passage below and answer the questions that follow.

Hope to have a baby

Artificial insemination of a woman with her husband's sperm allows an infertile couple to have a child without involving a third party. If the husband cannot produce sperms, then sperms from a donor may be used.

In vitro fertilization, or IVF, is used to help infertile women to give birth to a baby. The woman is first treated hormonally to super-ovulate. Ova are taken out from the woman's ovaries and mixed with a nutrient solution and semen for fertilization. The fertilized ova undergo cell divisions to form balls of cells. Several balls of cells are placed into the woman's uterus. If all go well, normal pregnancy and birth of a baby will follow.

(a) Suggest ***two*** possible causes of infertility in each sex.

(i) in female (2 marks)

(ii) in male (2 marks)

(b) (i) State the part where fertilization takes place naturally. (1 mark)

(ii) Describe the route of a sperm after reaching the vagina until it fertilizes an ovum. (2 marks)

(c) What is the aim of inducing super-ovulation? (1 mark)

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(d) Several balls of cells are placed in a female's uterus after IVF. What is the reason and what may be the possible results? (2 marks)

_____

_____

##

(a) (i) Any ***two*** of the following: (2)
- non-functioning of the ovaries (1)
- blockage of the oviducts (1)
- unsuccessful implantation (1)

(Or other reasonable answers)

(ii) Any ***two*** of the following: (2)
- low sperm count (1)
- poor sperm motility (1)

(Or other reasonable answers)

(b) (i) Oviducts (1)

(ii) It swims through the cervix and enters the uterus. (1)

It then swims to the oviducts and fertilizes an ovum there. (1)

(c) To allow several ova to become mature at the same time (1)

(d) Reason: to increase the chance of success of pregnancy (1)

Result: lead to multiple births (1)

(10 marks)

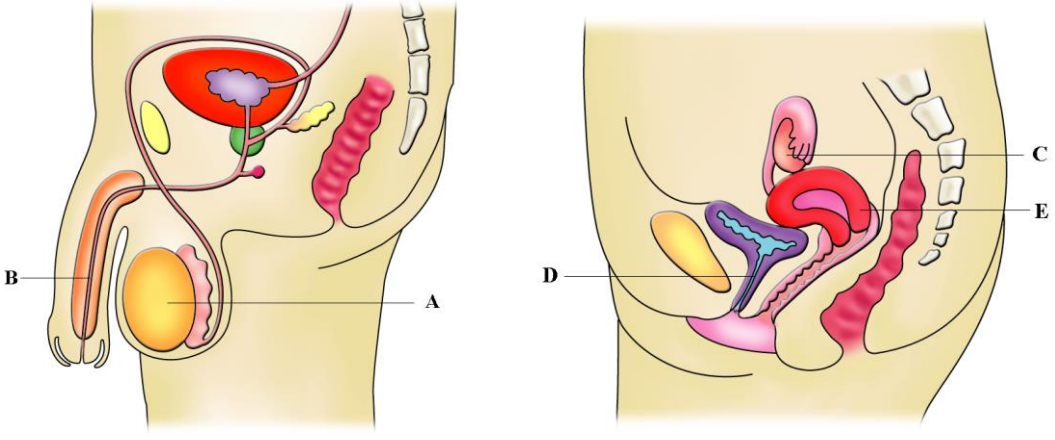
##

	Skill assessed	Marks
(a)(i)	Applying concepts	2
(a)(ii)	Applying concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2
(c)	Understanding basic concepts	1
(d)	Applying concepts	2

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|!|BCELC0140000011|!|

The following diagrams show the male and female reproductive systems in humans.



(a) (i) State the labelled structures that are involved in gamete production in both male and female. (1 mark)

(ii) State ***two*** adaptive features of the gametes produced by the male. (2 marks)

(iii) Describe how a female gamete is released from the structure you answered in (a)(i). (2 marks)

(b) (i) State the labelled structure that is involved in embryo development in the female body. (1 mark)

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(ii) Describe and explain the changes of this labelled structure after the release of the female gamete. (2 marks)

(iii) What will happen to this labelled structure if no fertilization occurs after the release of the female gamete? (1 mark)

##

(a) (i) Structures A and C (1)

(ii) Any ***two*** of the following: (2)

- It has tail to help it move. (1)
- Its midpiece is packed with mitochondria that supply energy for the movement of its tail. (1)
- It is haploid that can restore the diploid chromosome number in the zygote. (1)

(Or other reasonable answers)

(iii) The follicle gets mature and moves to the edge of the ovary. (1)

The mature follicle burst to release the ovum inside. (1)

(b) (i) E (1)

(ii) Structure E (uterine lining) will thicken. (1)

This can prepare the uterine lining for receiving an embryo. (1)

(iii) Structure E (uterine lining) will break down. (1)

(9 marks)

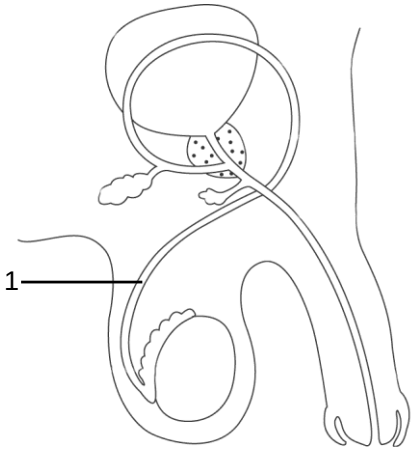
##

	Skill assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Understanding basic concepts	2
(a)(iii)	Understanding basic concepts	2
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	2
(b)(iii)	Understanding basic concepts	1

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|!|BCELB0140000012|!| ★

The diagram below shows the human male reproductive system.



(a) Explain why the secondary sexual characteristics of a man are not affected after structure 1 is tied and cut. (2 marks)

(b) A male condom can be used to achieve contraception.

(i) State how a male condom can be used to achieve contraception. (2 marks)

(ii) State another advantage of using male condom during sexual intercourse. (1 mark)

(c) (i) When a sperm starts penetrating the cell membrane of an egg, what will happen to the egg? (1 mark)

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(ii) Explain the significance of the change answered in (c) (i). (2 marks)

##

(a) The secondary sexual characteristics of a man are controlled by male sex hormones. (1)

Tying and cutting structure 1 will not affect their production in the testes. (1)

(b) (i) The ejaculated semen is trapped inside the condom and (1)

the sperms are prevented from entering the vagina and meeting the egg. (1)

(ii) It can prevent the transmission of sexually-transmitted diseases. (1)

(c) (i) The jelly coat of the egg thickens and hardens. (1)

(ii) It can prevent the entry of other sperms. (1)

This can prevent the chromosome number of the zygote from getting larger than normal / prevent the formation of abnormal embryo. (1)

(8 marks)

##

	Skill assessed	Marks
(a)	Applying concepts	2
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	2

|!|BCELB0140000013|!| ★

The diagram below shows the structure of a human sperm.



(a) Describe how the sperm can be transferred from a male to a female. (3 marks)

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(b) (i) State ***two*** structural differences between the sperm and a white blood cell. (2 marks)

(ii) Explain the importance of these differences in the sperm. (2 marks)

(c) Name structure A and state its function. (2 marks)

##

(a) The penis of the male becomes erect, (1)
and is inserted into the vagina of the female. (1)
Semen is ejaculated into the vagina when the muscles of epididymis and sperm ducts contract. (1)

(b) (i) The sperm has a haploid number of chromosome and the white blood cell has a diploid number of chromosome. (1)
The sperm has a tail and the white blood cell has not. (1)

(ii) The haploid sperm can help restore the normal diploid chromosome number during fertilization. (1)
The presence of tail in sperm can help sperm swim to the ovum. (1)

(c) Acrosome (1)
It contains enzyme that can be used to penetrate an ovum. (1)

(9 marks)

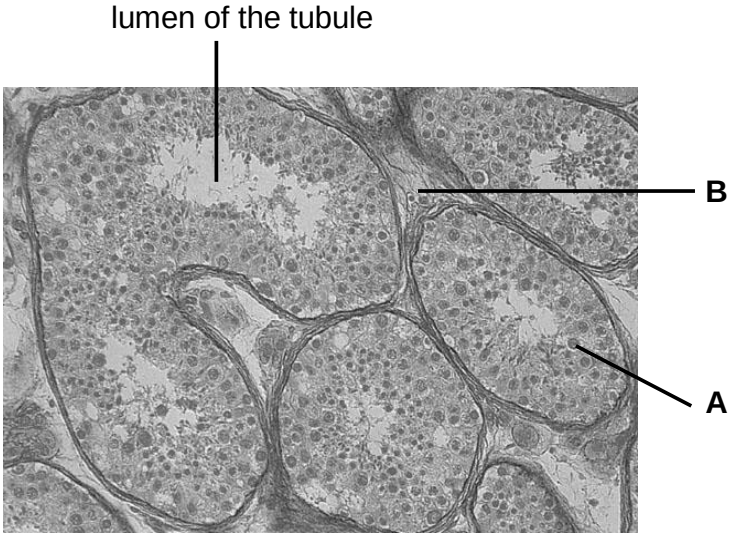
##

	Skill assessed	Marks
(a)	Understanding basic concepts	3
(b)(i)	Applying concepts	2
(b)(ii)	Understanding basic concepts	2
(c)	Analysing data	2

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|!|BCELB0140000014|!| ★

The photomicrograph below shows a section of the human testis which contains several seminiferous tubules.



(a) The diploid cells in region A produce sperms.

(i) Name the process by which sperms are formed. (1 mark)

_____

(ii) Draw a labelled diagram of a sperm in the space below. (3 marks)

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(b) Describe how the semen is formed and ejaculated. (3 marks)

(c) Some cells in region B can synthesize a male sex hormone.

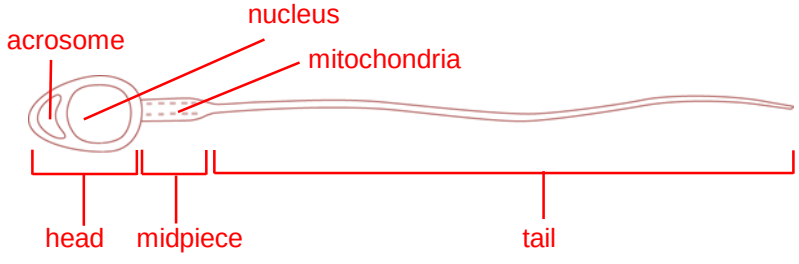
(i) State the chemical nature of this hormone. (1 mark)

(ii) State how this hormone can travel from the testes to the target organs. (1 mark)

##

(a) (i) Meiotic cell division (1)

(ii)



Structure of a sperm

Correct drawing (1)
Correct labels (Any ***three***) (1)
Correct title (1)

(b) The epididymis contracts and forces the sperms stored inside to move along the sperm ducts. (1)
Seminal fluid is secreted into the sperm ducts and mixed with sperms to form semen. (1)
The semen is finally discharged out of the body through the urethra of the erecting penis. (1)

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(c) (i) The chemical nature of male sex hormone is lipid / steroid. (1)

(ii) Male sex hormone is transferred to the target organ through the bloodstream. (1)

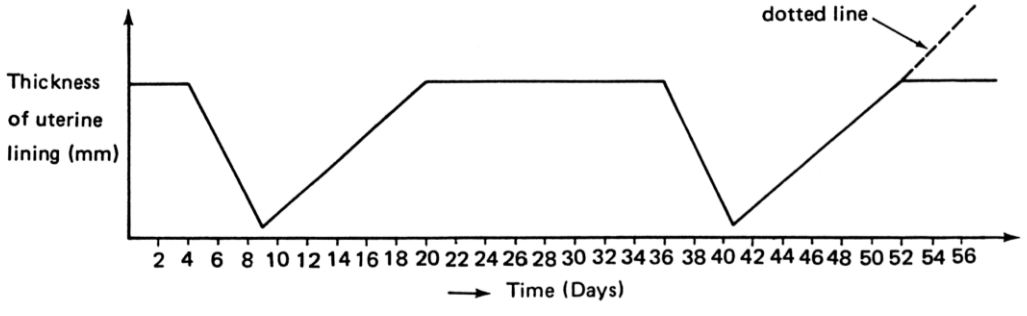
(9 marks)

##

	Skill assessed	Marks
(a)(i)	Understanding basic concepts	2
(a)(ii)	Presenting information	3
(b)	Understanding basic concepts	3
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	1

|!|BCELB0140000015|!| ★

The following graph shows the cyclic development of the uterine lining of a woman.



(a) How long does the menstrual cycle of this woman last? (1 mark)

(b) State the day(s) on which ovulation occurs. (1 mark)

(c) Explain how the uterine lining remains thickened from day 20 to day 36. (2 marks)

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(d) Explain the shape of the dotted line. (3 marks)

(e) Give ***one*** example for each of the following.

(i) An animal which carries out external fertilization (1 mark)

(ii) An animal which carries out internal fertilization with mainly external embryonic development (1 mark)

##

(a) 32 days (1)

(b) day 20 and 52 (1)

(c) After ovulation, the ruptured follicle becomes the yellow body. (1)
The yellow body produces oestrogen and progesterone and these two hormones cause the uterine lining to remain thickened. (1)

(d) The dotted line shows that the uterine lining continues to thicken. (1)
This is because fertilization occurs and the woman becomes pregnant. (1)
The thick uterine lining is ready for the implantation and the growth of the embryo. (1)

(e) (i) frogs / fish (1)
(ii) birds (1)

(9 marks)

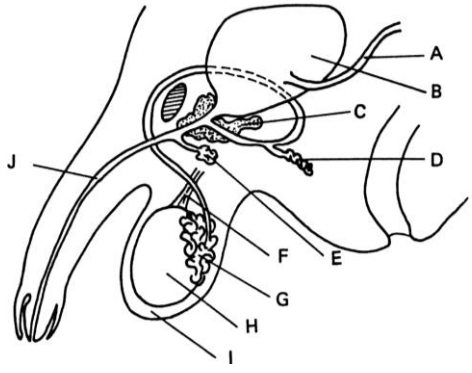
##

	Skill assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Understanding basic concepts	2
(d)	Applying concepts	3
(e)(i)	Understanding basic concepts	1
(e)(ii)	Understanding basic concepts	1

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|!|BCELC0140000016|!|

The following diagram shows the reproductive system of a man.



(a) State the labelled structure that is responsible for the temporary storage of sperms. (1 mark)

_____

(b) Name structure D and state its function. (2 marks)

_____

_____

(c) What would happen to a man if structure H is located inside the abdominal cavity? Explain. (2 marks)

_____

_____

(d) (i) State the labelled structure where male sex hormone is secreted. (1 mark)

_____

(ii) State ***one*** function of the male sex hormone. (1 mark)

_____

##

(a) Structure G (1)

(b) Seminal vesicle (1)

It secretes seminal fluid to activate and nourish sperms. (1)

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(c) He will have a low sperm count because (1)
the body temperature inside the abdominal cavity is too high for the production of sperms. (1)

(d) (i) Structure H (1)

(ii) It stimulates the development of the male reproductive system / secondary sexual characteristics. (1)

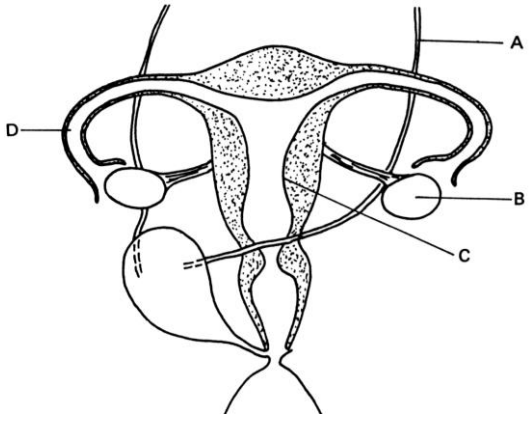
(7 marks)

##

	Skill assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	2
(c)	Applying concepts	2
(d)(i)	Analysing data	1
(d)(ii)	Understanding basic concepts	1

|!|BCELB0140000017|!| ★

The following diagram shows the structure of the female reproductive system.



(a) State the differences between the gametes produced in structure B and those by a man with reference to

(i) the sex chromosome present. (1 mark)

(ii) the amount of food reserve. (1 mark)

(b) How can the gametes produced by structure B be moved along structure D? (2 marks)

(c) Describe the changes of structure C

(i) during ovulation. (1 mark)

(ii) when the ovum is fertilized. (1 mark)

(iii)when the ovum is not fertilized. (1 mark)

##

(a) (i) Gametes produced in structure B contain X chromosome only while gametes produced by a man contain either X or Y chromosome. (1)

(ii) Gametes produced in structure B contains more food reserve than gametes produced by a man. (1)

(b) Structure D (oviduct) is lined with ciliated epithelium. (1)

The beating action of cilia and muscular contractions of the oviduct wall help move an ovum (gamete produced by structure B) along the oviduct. (1)

(c) (i) Structure C becomes thicker. (1)

(ii) When the ovum is fertilized, structure C becomes much thicker. (1)

(iii)When the ovum is not fertilized, the thickened uterine lining will break down. (1)

(7 marks)

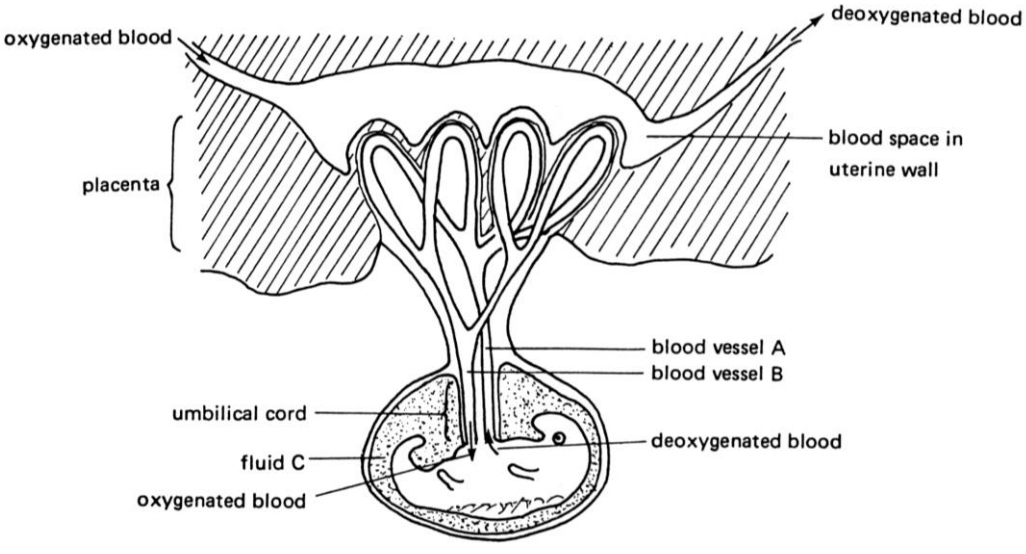
##

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	Skill assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	1
(c)(iii)	Understanding basic concepts	1

|!|BCELB0140000018|!| ★

The diagram below shows part of the uterus of a pregnant woman.



(a) State ***three*** differences in the composition of the blood in blood vessels A and B. (3 marks)

(b) Besides acting as the site for the exchange of substances between the mother and the embryo, state ***two*** other functions of the placenta. (2 marks)

(c) State ***two*** significance of the separation of the maternal blood and the embryo's blood in the placenta. (2 marks)

(d) Explain why pregnant women should not smoke and drink alcohol. (2 marks)

##

(a) Any ***three*** of the following comparisons: (1 mark each) (3)

Blood in the umbilical artery	Blood in the umbilical vein
- less oxygen	- more oxygen
- less nutrients (e.g. glucose, amino acids)	- more nutrients (e.g. glucose, amino acids)
- more urea	- less urea
- more carbon dioxide	- less carbon dioxide

(Or other reasonable answers)

(b) Separates the maternal blood and the embryo's blood (1)

Secretes hormones to maintain the thickness of the uterine lining (1)

(c) This can prevent the high blood pressure of the maternal blood from breaking the delicate embryonic blood vessels. (1)

This can prevent pathogens from entering the embryo. (1)

(d) The smoke contains nicotine and carcinogenic substances. (1)

Nicotine, carcinogenic substances and alcohol can enter the embryo's blood through the placenta and cause harm to the embryo / affect the growth of the embryo. (1)

(9 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)	Applying concepts	2

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